

Bhargi Bhalara

bhargibhalara03@gmail.com | 9735806664 | San Jose, California | linkedin.com/in/bhargibhalara

SUMMARY

An ardent engineer with over eight years of experience in integrating FPGA-based solutions into complex systems, showcasing expertise in RTL Designs for SoC/FPGAs/ASIC, VLSI, digital and analog circuit design, signal processing, and embedded systems. Skilled in end-to-end FPGA design flow with seamless post-design integration, validation, and debugging efforts. Proven ability to enhance system architecture with meticulous documentation and automating test processes, significantly improving efficiency and accuracy.

EXPERIENCE

Electrical Systems Engineer

Reflexion Medical, Inc.

Jun 2021 - May 2025

- Implemented end-to-end FPGA design flow including Synthesis, Implementation, Place and Route, and Bitstream Generation for Xilinx Ultrascale FPGAs using Vivado design suite, achieving a 20% reduction in design cycle time.
- Improved the existing state machine to reduce the delay by 3-4 clock cycles, enhancing system performance by 15%.
- Automated over 50 regression test cases using Python and standard verification methodologies, resulting in a 30% decrease in manual testing time.
- Developed embedded test programs and prepared validation protocols for testing bare metal systems, improving validation accuracy by 25%.
- Debugged High-Speed Serial Interfaces such as PCIe and CAN Bus using protocol analyzers, facilitating a 40% increase in data throughput.
- Initiated engineering change orders and supported the design transfer process for updated FPGA designs, ensuring compliance with industry standards.

Electrical Engineer

Mirion Technologies (Capintec) Inc.

Apr 2017 - Jun 2021

- Remodeled RTL design logic for FPGA/CPLD used in Multi-Channel Analyzers and related PCB layout, leading to a 15% increase in processing speed.
- Supported design changes for electronic components like Microcontroller, SDRAM, PMIC, and similar CMOS devices in PCBs, ensuring timely delivery of designs to production and reducing time-to-market by 20%.
- Delivered design changes to production efficiently, following the design control process compliant with ISO-13485 and other regulatory bodies like IEC-60601, achieving zero non-compliance issues during audits.
- Led hardware validation for Dose Calibrators, Thyroid Uptake Systems, and Radiation Measurement Devices, ensuring accuracy and reliability in measurements with a 98% success rate.

Electronics Engineer

Reliance Electronics Inc.

Jan 2017 - Sep 2017

- Designed and validated printed circuit boards (PCB) using custom test fixtures, ensuring correct functionality post-manufacture and reducing error rates by 30%.
- Structured the BOM for circuit designs by selecting appropriate parts and updating designs due to EOL/obsolescence, ensuring continuity in production and minimizing delays by 25%.

Design Engineer

Scalable Systems Research Labs

Feb 2016 - Dec 2016

- Structured IEEE Floating Point Numbers Multiplier in Verilog for the Floating Point Unit used in RISC-V CPU, improving computational efficiency by 20%.
- Reduced delay on carry propagation by implementing vertical slice compression technique, hence increasing the speed of the processor by 15%.

FPGA Intern

Finisar Corporation

Jun 2016 - Sep 2016

- Developed comprehensive Verilog code for 10G Ethernet transceivers on Xilinx Kintex 7 FPGA, enhancing data transmission rates by 25%.
- Enhanced power efficiency by optimizing the state machine for Microsemi FPGA, resulting in a 10% reduction in power consumption.

Instructional Student Assistant

San Jose State University

Aug 2014 - Dec 2014

- Assisted students with their Advanced Digital Logic Course to teach digital design concepts like combinational and sequential circuits, Logic Reduction Techniques, and simulation of VHDL code in ModelSim, contributing to a 90% student pass rate.

EDUCATION

Masters of Science in Electrical Engineering

San Jose State University, San Jose

Dec 2015

GPA: 3.4

Bachelor of Engineering

Sarvajani College of Engineering and Technology

May 2013

SKILLS

AMBA (AHB, APB, ASB, AXI), ASIC, C, C++, Chipscope/ModelSim/Questasim, JIRA, JTAG, Lab Equipment and tools, Linux, Mentor Graphics PADS, Shell, SoC, System Verilog, USB, Verilog, CPU Architecture, GIT, HDMI, Intel Quartus II/Prime, Jenkins, Microchip Libero SoC, PCIe/PCI, Python, QNX, QuestaSim, RTOS, VHDL, Vitis HLS, VUnit verification framework, Xilinx ISE Design Suite, Xilinx Vivado Design Suite